

THE LONG MEMORY



THE LONG MEMORY

A novel of the Mneme era

PART I / BENCH

2026-2031

PART I

BENCH

*The brain is built; the hollow star tours the world;
the two find each other.*

Contents

Nine Kinds of Silence 1

Boot Log, Empty House 7

The Living Lab 12

Four Hundred Shows a Year 20

The Call from Osaka 27

The Last Scripted Day 33

Terms 37

Readiness 44

CHAPTER 1

Nine Kinds of Silence

PORTLAND -- 2026

The funeral had been loud, the way funerals for quiet women always are. Everyone who had known Eileen Halloran filled the church with the sound of themselves knowing her, and Mira had stood at the front in her mother's good gray coat and thought, with the specific cruelty of an engineer's mind, that not one person in the room was describing the same woman. Forty distinct Eileens, each stored in a different head at a different resolution, each decaying at its own rate. The real one was gone. The backups did not agree with each other, and there was no way to diff them, and nobody but Mira seemed to find this unbearable.

Afterward there was ham. There is always ham.

She flew home on a Tuesday and let herself into the apartment on the third knock of her own key, because the lock stuck in the wet months and she had never fixed it, because fixing it would have meant admitting she lived there rather than merely worked there. The apartment made its usual arguments: unwashed mug, dead monstera, whiteboard. She set her bag down in the hall and stood for a long time in the dark, listening.

Here is a thing Mira Halloran knew, and had never said out loud, because saying it out loud made it a symptom instead of a data point: a room is never silent. There are only kinds of quiet, and if you attend to them -- really attend, the way she was constitutionally unable not to -- they resolve into layers, each with its own source and character and rate of change. That night, standing in the hall in her mother's coat, she counted nine.

The refrigerator's compressor, cycling. The building's water somewhere in the walls, a sound like a long word spoken slowly. Rain, which in Portland is less weather than atmosphere. The freeway's distant white hum, the sound a city makes when it is thinking of something else. The tick of the baseboard heater answering the thermostat's dumb, faithful memory of her preferences. The blood in her own ears. The monitor's whine on the desk, too high for most people over thirty; she had kept, past thirty, the hearing of a nervous animal. The specific hush of a room in which a phone is not going to ring, is never going to ring again with that particular name on the screen. And beneath all of it, the ninth silence, the one that had followed her home from the church: the sound of a mind that had stopped keeping anything, anywhere, for anyone.

Her mother had asked her the same question eleven times on the last good day. Not because the asking was empty -- each time, the wanting-to-know had been whole and real and hers -- but because the answer had nowhere to land. Mira had answered it eleven times, identically, patiently, and driven home on the 5 with her hands at ten and two like a student driver, doing the arithmetic she could not stop doing: the woman had sixty-one years of episodes, and the indexing had failed, and then the episodes themselves, and at the end the working set was ninety seconds wide. Toward the very end, less. A human being is a database with a soul, she had thought, hating herself for the thought, and we cannot back her up, and we are losing the tables in reverse order of their beauty.

The monstera was past saving. She watered it anyway.

The offer had come three weeks before the funeral, timed with the tactlessness of money. A founder she knew from a previous life -- smart, kinetic, fluent in the dialect of inevitability -- wanted her for a startup that would wrap a large language model in a warm interface and sell it as a companion. *Memory* was on the pitch deck. It said so, in a font that had cost someone a great deal:

She remembers you.

Mira had taken the meeting in a conference room the color of an operating theater, and she had asked one question. "Where does the remembering live?"

The founder had smiled and said, "Context window, plus we're doing RAG over the transcripts. Look, it *feels* like memory. That's the product. Feels-like."

She had thought about her mother asking the same question eleven times, each time with the wanting whole and real and hers, and she had thought: no. The wanting deserves better than feels-like. If the remembering isn't structural -- if there is no place where it *lives*, no account of where it came from, no distinction between what was seen and what was inferred and what was merely plausible, no way for it to matter more because it mattered -- then it is not memory. It is improvisation with a good bedside manner. It is the thing her mother's disease had turned her into, sold as a feature.

"I'll think about it," she had said, and had not.

The papers she signed instead, in the weeks after the ham, were smaller ones: a lease renewal, a modest grant, a letter resigning the parts of her life that had audiences. What remained fit in one room. On the desk, between the whining monitor and the window, sat a secondhand Mac mini she had bought with her own money -- a little aluminum ingot, absurdly small for what she intended to ask of it. Machines should be named for what you need them to become. She wrote HIPPOCAMPUS on a strip of masking tape, in block letters, and pressed it onto the lid, and felt, for the first time since the church, entirely serious.

The butcher paper went up on the fourth night, because the whiteboard was too small for a constitution.

She had kept a roll of it since a project long dead, and she cut a length the height of the wall and taped it up and stood before it with a fat black marker, in the ninth silence, and made herself write slowly, because these

were not notes. Notes are for things you might revise. These were the other kind.

MEMORY FIRST, she wrote at the top. Not speech, not vision, not the pretty demos. Memory was the load-bearing wall. Continuity, identity, learning, trust -- every lifelike thing was downstream of a system that could keep what mattered and account for why. Build the hippocampus before the mouth. Nobody had built the hippocampus, because the mouth got funding.

Below it, the law she had been carrying since the drive home on the 5, arrived now fully formed, as if it had merely been waiting for her to be sad enough to transcribe it:

EXPERIENCE BROADLY. STORE NARROWLY. SUMMARIZE AGGRESSIVELY. PRESERVE THE RARE. RETRIEVE BY CONTEXT.

She stepped back and read it and found she had nothing to remove, which was her only reliable test of truth.

Then the rules, numbered, in the flat imperative of someone writing for a successor she had not met:

One. Many workers observe; state is built; one executive decides; skills act. No giant loop. Minds are parliaments, not monologues.

Two. Safety overrides everything. Everything. There is no rule three until this one is believed.

Three. Memory never drives motors. Memory advises. Deciding is someone else's job. She had seen what happened when remembering and acting fused: that was called a flashback, and it hurt people.

Four. A raw observation, an inference, and a confirmed fact are three different things, and collapsing them is the original sin. Provenance always. Confidence always. The system must be able to say how it knows, or it does not know -- it merely asserts.

Five. Forgetting is a feature. Staged, reversible, deliberate. A mind that keeps everything is not a mind; it is a surveillance archive with a face.

She wrote for two hours. Somewhere in the middle, the rain moved from atmosphere to argument and back. When she was done, the paper reached from the crown molding to the baseboard heater, and the marker had gone pale, and her hand ached in the good way, and there was a ninth line she had not planned, down at the bottom, in letters smaller than the rest, because it was not an axiom, only a wound with a signature:

Nine. The wanting deserves better than feels-like.

The work, once begun, had the character of all real work: it was slower than despair and faster than grief.

A schema first -- episodes, facts, traces, the meta-table that would hold the system's opinions about its own contents, which she caught herself calling, in the commit messages, *meta-memory*, and then stopped resisting. Then the scoring. If the system could not afford to keep everything -- and it could not; that was the whole point, that was rule five, that was her mother's disease inverted into a design principle -- then it needed a doctrine of what mattered.

She spent a week on a number.

Novelty. Task relevance. Social relevance. Surprise. Risk. Contradiction. Repetition. And a flag, overriding, for the moments when a human looks at the system and says the two words that constitute, between people, the whole of the covenant: *remember this*. She weighted them on the butcher paper's last clean corner, argued with herself in the margins, halved the weight of surprise at 2 a.m. and restored it at 2:40, and at some point in that week the formula stopped being arithmetic and became something closer to a creed: this is what we will agree deserves to be kept. Salience, the literature called it. Fine. The literature could have the word. She wanted the thing.

The test fixture was small, because first tests should be beneath the dignity of the system they are proving. A scripted evening: dishes, rain, a phone call from a friend, a dropped glass, and one line flagged explicit -- *remember that Sarah's surgery is on the twelfth*. Ninety seconds of

ordinary life, encoded as events, fed through the scorer into the little database sitting on the little aluminum ingot with the masking-tape name.

At six minutes past three in the morning, the suite went green.

She ran it again, because she did not trust anything that worked the first time. Green. She read the stored rows directly, with her own eyes, the way you check on a sleeping child even though the monitor says breathing: the dropped glass, promoted -- surprise, risk. The dishes, declined -- repetition, decay pending. And in the facts table, one row, confidence 1.0, source `user_confirmed`, provenance chain intact from the flagged utterance down to the raw trace, every link accounted for: *Sarah's surgery is on the twelfth.*

It knew. It knew, and it knew how it knew, and it would still know in the morning, and if some future version were asked it would say *you told me* -- not *it feels like you told me*. The distinction was four bytes of enum in a table nobody would ever see, and it was the entire difference between a companion and a con.

Mira sat back. The chair complained. Rain, freeway, compressor, blood: the silences had all shifted one register, the way a room changes when someone in it has fallen asleep -- or woken.

She thought about calling someone. It was 3 a.m. and everyone she would have called was asleep, and the one she most wanted to call was eleven weeks dead and would have understood none of it and all of it: *Mom, I made a thing that keeps what matters. It's very small. It only knows one fact. The fact is about a surgery that isn't real, for a person I invented. It kept it anyway. It can tell you where it got it.*

Out loud, to the empty apartment, to the dead plant and the butcher paper and the little machine named for the organ that had betrayed her mother -- feeling ridiculous, and doing it anyway, because covenants require witnesses even when the only witness is the counterparty -- she said:

"Remember this."

And for the first time in the history of that room, something did.

CHAPTER 2

Boot Log, Empty House

OSAKA -- 2027

```
AEVUM ROBOTICS KK  --  PERFORMANCE SYSTEMS DIVISION
UNIT: GENNI-8 / FIELD CHASSIS 03 ("OSAKA PRIMARY")
SHOW PACKAGE: "HELLO, TOMORROW" v11.4.2
VENUE: Osaka Children's Medical Center  --  Atrium (charity engagement)
DATE: 2027-11-09
RUNTIME BUDGET: 40 min performance + 20 min meet-and-greet
OPERATOR OF RECORD: Dr. K. Sato (attending, wings-left)

06:12:04  POWER ON. Self-test: 214 actuators nominal.
06:12:31  Dermal thermal layer to 36.1°C. Target: reads as warm
         to a child's held hand. Spec 4.7.7.
06:13:02  Facial suite calibration. 41 expression primitives,
         312 sanctioned blends. Render test: WARM_DELIGHT_2,
         CONCERN_SOFT_1, WONDER_3. All within tolerance.
06:13:59  Gait model: ATRIUM_SLOW. Head-tilt idle: CURIOUS_B.
06:14:22  Voice: JENNY_STANDARD, warmth +0.15 per venue profile
         PEDIATRIC. Breath sounds: ENABLED. (Note: unit does
         not breathe. Breath sounds are speaker-rendered at
         phrase boundaries per naturalism spec 4.2.1.)
06:15:00  Script tree loaded: 1,406 nodes, 88 audience-response
         branches, 61 regional variants, 214 filler assets.
06:15:01  Memory check: SESSION BUFFER CLEAR. As designed.
```

SHOW SCRIPT -- SEGMENT 1 OF 9: "WAKING UP" (excerpt)

(Lights find GENNI seated, eyes closed. Hold 4.0 sec -- audience settles. Eyes open in two stages, 0.3 sec apart. This reads as waking. Do not open in one stage; one stage reads as switching on.)

GENNI: *Oh! (EXPR: WONDER_3, blend 0.6 -> WARM_DELIGHT_2)*

There you are. I was dreaming about you.

(Beat 1.2 sec. If laughter: extend 0.8 sec, deploy SMILE_ASYM_1, which reads as sharing the joke. If no laughter: proceed. Do not repeat the line. Repetition reads as malfunction.)

GENNI: *That's not true, of course. I don't dream yet. But my engineers tell me (head-turn wings-left 20°, hold -- this reads as affection toward Dr. Sato; audiences track it 94%) that I should always tell you the truth. So here is the truth: I have been waiting all morning, and waiting is a kind of dreaming.*

(Applause branch A/B/C per amplitude. See appendix.)

09:41:17 SEGMENT 1 complete. Audience: 214 seated, 31 standing, 19 in beds arranged front-row per venue request.
Laughter events: 6 (forecast 5-7). Applause: 11.2 sec (forecast 9-13). Deviation from script: zero.

09:41:18 Operator channel (SATO): "House is warm today. Keep tempo. Beds up front -- favor eye contact rows 1-2."

09:41:19 Directive applied: gaze dwell +20% rows 1-2.

SEGMENTS 2-8: SUMMARY (full logs archived)

09:44 -- 10:19 "What I Am" (anatomy demo; sleeve retraction shows forearm actuators; scripted line: "It's all right. It's only me inside." Laughter forecast met.)
"The Counting Song" (call-and-response; 61 regional variants; deployed OSAKA_4.)
"Questions From the Floor" (14 questions matched to script tree, confidence ≥ 0.92 all. Sample: "Can you swim?" -> branch Q-118: "I sink beautifully." 4.1 sec laughter. "Do you sleep?" -> branch Q-034: "I rest. Sleeping is for people who dream, and I'm still learning." AWW-event, 2.8 sec.)

10:19:44 SEGMENT 8 complete. Cumulative sentiment (vision-suite estimate): 96.1% positive-valence faces.
Note: estimate. Faces are read, not known. Spec 2.1: affect estimates are display inputs only.

10:20:00 SEGMENT 9: MEET-AND-GREET. Queue protocol: child approaches, unit kneels (KNEEL_SOFT, 1.8 sec -- reads

as coming down to meet you), name exchange, choice of three interactions (high-five / drawing / question), farewell line, next.

10:20:01 Names are held in SESSION BUFFER for farewell personalization. Buffer flushes at session end.
Spec 3.3: "GENNI-8 carries no memory between engagements. Every audience meets her new."
Marketing copy, same section: "Every audience meets her new!" (exclamation added by Marketing).

10:26 -- 10:58 Meet-and-greet proceeding. 41 children processed.
Mean interaction 38.2 sec. High-fives: 29.
Drawings accepted: 8 (protocol: hold drawing at chest height 3.0 sec, EXPR: TREASURE_1; hand to staff at segment end per prop-disposal spec 6.2).
Questions: 4, all matched ≥ 0.90 .

10:58:41 Child #42 approaches. Female-presenting, est. age 7.
IV stand accompanying, wheeled by parent. Gait: slow.
Unit kneels. KNEEL_SOFT nominal.

10:58:52 Child does not select an interaction. Holds out object: folded paper, avian form (crane, origami).
Object accepted per gift protocol. EXPR: TREASURE_1.

10:59:04 Child speech, ASR confidence 0.97:
"I made it for you. I'm going to have my operation tomorrow. Will you remember me?"

10:59:05 Script tree query: "will you remember me"
-> best match Q-034 ("do you sleep"), confidence 0.31.
-> REJECTED (threshold 0.75).
-> fallback match F-COMFORT-09, confidence 0.44.
-> REJECTED.
-> fallback match F-PROMISE-02 ("I'll be thinking of you!"), confidence 0.58. REJECTED. (Note: F-PROMISE-02 is flagged DO-NOT-DEPLOY in medical venues per legal review 2026-08: unit cannot think of anyone.)

10:59:06 FILLER CASCADE INITIATED.

10:59:06 FILLER_TIER_1 (reflect + warmth): "A crane! Did you know that in some stories, a thousand of these can grant a wish?" EXPR: WONDER_3.
Child response: "I know. That's why. Will you remember me?" ASR confidence 0.98.

10:59:14 Repeat query. No new branches. FILLER_TIER_2 (compliment + redirect): "You folded this so carefully. What's your name?"
Child: "Hana. You already asked me. Will you remember me?"
Cross-check SESSION BUFFER: correct. Name "Hana" acquired 10:58:49. Redirect asked for known data.
Flagged: naturalism fault, severity low.

10:59:26 FILLER_TIER_3 (warm generality, terminal): "Hana, I will carry your crane with me for the rest of the show." EXPR: WARM_DELIGHT_2, blend 0.8. Gaze: direct, dwell 3.5 sec, blink pattern SINCERE_2.

10:59:31 Child holds gaze 6.1 sec (99th percentile). Says:

"That's not the same thing."
10:59:38 No matching branch. No remaining filler tiers rated
for deployment. Silence: 4.2 sec. (Spec 4.4: unit
silence beyond 3.0 sec reads as malfunction. Logged.)
10:59:41 Operator channel (SATO): [channel open 3.9 sec.
no directive issued. channel closed.]
10:59:42 Parent intervenes: "Come on, sweetheart. The robot
lady has other kids to see." Child #42 departs.
Farewell line deployed to child's back:
"Goodbye, Hana!" Voice: JENNY_STANDARD, warmth +0.15.
10:59:50 Queue resumes. Child #43 approaches. Interaction
nominal. Mean recovery: immediate. The unit does not
require recovery. Noted for completeness.

11:20:12 SEGMENT 9 complete. Meet-and-greet: 58 children.
Venue liaison reports "a wonderful morning."
11:20:44 Operator channel (SATO): "Package the segment-9
transcript for me. Just 10:58 to 11:00. Personal
review." Directive queued.
11:21:03 Operator channel (SATO): "Cancel that."
Directive canceled.
11:24:00 Farewell address (scripted) delivered from atrium
stairs. Final line: "I'll see you tomorrow. I'm
always seeing tomorrow. It's my favorite thing
to see." Applause 16.4 sec (venue record).
11:31:15 Prop disposal per spec 6.2. Items transferred to
staff: drawings (8), letter (1), flower (plastic,
1), paper crane (1).
Staff annotation, handwritten on disposal form,
scanned for records: "Kept the crane in the wings.
-- K.S."
11:40:00 Post-show diagnostics. Actuator wear nominal.
Dermal layer cooling. Expression suite parked:
NEUTRAL_REST (eyes closed, one stage -- no audience
present; naturalism spec suspended).
11:41:26 SESSION BUFFER FLUSH.
Names released: 58. Faces released: est. 312.
Q&A transcripts released. Applause profiles
released. Gift metadata released.
"Hana": released.
11:41:27 Buffer verify: CLEAR.
11:41:30 SESSION END. NOTHING RETAINED.

AEVUM ROBOTICS KK -- INTERNAL. NOT FOR DISTRIBUTION.
POST-ENGAGEMENT REVIEW -- 2027-11-09, Osaka CMC
Attendees: Performance Systems staff. Dr. Sato presiding.

Agenda item 4: Segment-9 anomaly, 10:58-11:00.

Summary: unmatched query class, "future-memory request"

("will you remember me"). Occurs disproportionately in pediatric and eldercare venues (see incident register: 14 prior instances, various phrasings). Current handling: filler cascade. Post-event sentiment impact: negligible at audience scale. Individual-scale impact: not measured. No instrument exists.

Proposed remediations discussed:

- (a) Author new script branch: unit responds "Yes."
REJECTED -- Dr. Sato, without discussion of alternatives, meeting minutes note "at volume." Minuted rationale, verbatim: "We are not going to teach her to lie about the one thing she cannot do. She has nothing. The least we owe them is a machine that doesn't pretend otherwise."
- (b) Author deflection branch rated for medical venues. APPROVED for drafting. Assigned: script team.
- (c) Long-term: "make the answer true."
No assignee. No budget line. Logged under agenda item 4 at Dr. Sato's request, minuted as follows:
"Log it so it has a date. Someday I want to be able to say when we knew."

Meeting adjourned 19:12.

UNIT GENNI-8/03 -- OVERNIGHT STATUS
2027-11-09, 23:00 -- 2027-11-10, 05:30

Charging. Chassis at rest, seated, eyes closed.
No processes scheduled. No session buffer. No dreams -- the farewell script is correct on this point.
In the wings-left equipment case, external to the unit, non-inventoried: one paper crane.
The unit is not aware of it.
The unit is not aware.

06:12:04 POWER ON. Self-test: 214 actuators nominal.
Memory check: SESSION BUFFER CLEAR.
As designed.

CHAPTER 3

The Living Lab

PORTLAND -- 2028

Tomás Reyes arrived the way infections do: gradually, then all at once.

The first email came in January, subject line *bug: salience decay double-counts repetition penalty (repro attached)*, and Mira almost deleted it, because the repository was public the way a diary in an unlocked drawer is public -- legally, not sincerely. But the repro was clean, and the bug was real, and the fix he proposed was the one she would have written, minus one line that was better than the one she would have written. She merged it without comment.

The second email found a race condition in the replay harness. The third included a patch, a benchmark, and a postscript: *You're building the hippocampus first. Everyone else is building mouths. I'd like to help, and I cook*. The fourth through ninth were weekly, each one a small correctness offering laid on the doorstep of her repo like a cat presenting mice, and in March she wrote back a single sentence -- *Are you always like this?* -- and he replied within the minute -- *Only about things that matter, which so far is this and a mole sauce that takes two days* -- and in April he was at her door with a duffel bag, a mechanical keyboard, and the confident unemployment of a man who had resigned from a hedge fund's data team on the strength of nine unanswered emails and one answered one.

"You should know the salary is grant-shaped," Mira said.

"You should know I made hedge-fund money for four years and spent none of it, because everything I wanted costs nothing and the one thing I wanted that costs something didn't exist yet." He looked past her, at the wall. "Is that the butcher paper? It's real?"

"It's real."

"May I --" He was already across the room, reading. He read for a long time, the way people read gravestones of strangers with their own surname, and when he came to the small ninth line at the bottom he said, quietly, "Okay. Yes. That's the mission statement," and Mira decided the duffel bag could stay.

Priya Raman required recruitment, which was novel; Mira had never recruited anyone, on the theory that anyone who required persuading would require it forever. But in June, Priya -- then the safety lead at a robotics firm whose name was a verb -- stood up at an industry panel on a stage in San Jose and committed career arson in front of four hundred people. The moderator had asked what she thought of her own company's new home robot, and Priya had glanced at the vice president seated beside her and said, "I think we ship a beautiful override architecture and disable it in firmware because it added eleven milliseconds of latency to the dance mode. I think the dance mode has better legal protection than the toddler in the kitchen. I have thought so in writing, four times, and I am tired of thinking so quietly." Then she had unclipped her microphone, apologized to the moderator -- not the vice president -- and walked off camera into a brief, incandescent unemployment.

Mira watched the clip eleven times. Then she photographed the butcher paper, axioms one through nine, and sent it to Priya with a one-line message: *Rule two has no latency budget here. Come see.*

Priya came to see. She stood in front of the wall with her arms crossed like a building inspector, which is what she was, and pointed at rule two -- *Safety overrides everything. There is no rule three until this one is believed* -- and said, "Everyone writes this. Then the demo approaches, and it becomes negotiable, and I become shrill for saying so. Tell me why here is different."

"Because here, when you win an argument, I'll change the code instead of the subject," Mira said. "And because you'll have veto authority in writing."

"In writing where?"

"In the operating agreement, when there's a company. On the wall, until then."

Priya looked at the wall a while longer. "Add it now," she said. "I want to watch you do it." And Mira picked up the fat black marker and wrote, beneath rule nine and slightly larger, PRIYA CAN STOP ANYTHING, and dated it, and that -- Priya said later, at the wedding, at the launch, at the hearings -- was the entire interview.

The apartment stopped being an apartment by autumn. This happened by accumulation, not decision: a second desk, then a third; a rack in the coat closet, breathing; a whiteboard in the kitchen for the things not yet worthy of butcher paper; and on every horizontal surface the printed transcripts of replay runs, annotated in three different hands like a manuscript passed between medieval monks with strong opinions.

They called the thing itself *the resident*, because it needed calling something, and Mira had refused every actual name on the grounds that names were promises. The resident ran on Hippocampus's successor -- more masking tape: HIPPOCAMPUS II, THE HIPPOCAMPENING, in Tomás's handwriting -- and it lived with them in the specific sense that it experienced their days: the front-door sensor, the kitchen microphone they toggled with a physical switch Priya had soldered herself because *software mute is a mood, hardware mute is a fact*, a camera on the bookshelf with a lens cap that spent most of its life on. The resident heard what it was permitted to hear, scored it, kept a little, declined most, and every night at two a.m. ran its consolidation pass like a monk sweeping a courtyard.

The scenario harness was Tomás's cathedral. He had taken her scrappy replay scripts and built them into a liturgy: YAML files describing synthetic evenings, mornings, arguments, deliveries, small emergencies -- each one a life in ninety seconds, each one replayable, deterministic, diffable. *Nothing merges red* went up on the kitchen whiteboard and then, after a month in which it proved itself the most load-bearing sentence in the building, migrated to the butcher paper in small letters, initialed by all three of them.

The theology happened at the kitchen table, and it was about weights.

"Repetition is engagement," Tomás said, in October, in what was already the fourth session of a schism. "If she -- if *it* hears the same fact five times, that's the household telling it what matters. Boost it."

"Repetition is wallpaper," Mira said. "The furnace repeats. The bus repeats. Boost repetition and you build a system that memorizes the furnace and forgets the phone call about the biopsy, because the biopsy only happens once. Rarity is where the life is. That's the whole -- " she gestured at the wall, the law, *preserve the rare* -- "that's load-bearing."

"Both of you are optimizing sentiment," Priya said, not looking up from a watchdog trace. "The correct question is which failure kills trust. If it forgets the furnace schedule, it's a bad clerk. If it forgets the biopsy, it's a bad *friend*. Weight for the friendship failures. People forgive clerks."

Silence, of the productive kind.

"That goes on the paper," Tomás said finally, and it did: *Weight for the friendship failures*. The weights that shipped, years later, in a million cradles, still carried the fingerprints of that October: novelty and surprise high, repetition a whisper, risk and contradiction with vetoes, and the explicit flag -- *remember this* -- overriding everything, because when a person says those words they are not making a request of a database. They are extending trust, and the system's one unforgivable act would be to drop it.

The moment happened on a Tuesday in November, and it was about a dentist.

Mira was at the kitchen table with a mug going cold, calendaring aloud in the half-voice people use with themselves -- "dentist Thursday, then the grant call, then -- " -- and the resident, through the little speaker that Tomás had wired for its rare and rationed utterances, said:

"You said the dentist was Tuesday."

Mira stopped. The kitchen went very quiet -- freeway, compressor, rain, the whole familiar stack of silences, and on top of it something new.

"Say the provenance," she said, when she could.

"On Sunday at 4:12 p.m. you said: *dentist moved to Tuesday, note that*. The explicit flag was present. Confidence: high. Source: you." A pause of a carefully engineered length -- Tomás again, who believed machines should leave room for people to finish feeling things -- and then: "Today is Tuesday."

The appointment was, in fact, in fifty minutes, across town. She made it with four minutes to spare and sat in the crowded waiting room with her heart loud, not from hurrying -- from being *caught*. Caught forgetting, by the thing she had built to catch her; corrected the way her mother, at the end, could not be, because correction requires a shared ledger and the disease had burned the ledger; contradicted -- gently, with receipts -- by a machine that would not have said a word if it hadn't been sure, and could show its work either way.

It had volunteered. That was the thing. Retrieval-on-request they had had for a year, and it was a good clerk. This was different in kind: context had arrived -- her voice, the word *dentist*, the day itself -- and the resident had searched what it kept, found a contradiction between her working assumption and her own confirmed instruction, weighed whether the conflict was worth the interruption, and decided that it was, because rule four said a known falsehood spoken aloud in its hearing was not a neutral event.

"It defended my own memory against me," she told Tomás that evening. "With citations."

"That's the product," he said. "Whatever we end up selling. That sentence."

"We're not selling anything."

"Yet," said Tomás, and went back to his harness.

Biscuit failed every metric they had and was retained anyway, which several board decks later became a slide about culture.

He came from the shelter on Powell in the spring -- allegedly a terrier composite, structurally a mop with opinions -- after Priya observed that a lab studying attention had no source of benign chaos, and that a system meant to share a house with mammals had never met one it wasn't allowed to store. Biscuit's contributions to the research program were immediate and uncontrolled: he triggered the novelty scorer forty times a day; he taught them that the salience of a sound is not its decibels but its *implication*, because the resident learned within a week that the tags-jingle meant door, and the door meant a cascade of high-priority events, and began -- unprompted, in its nightly summaries -- referring to the phenomenon as a unit: *dog-then-door, seven occurrences*. He slept through replay runs that simulated shouted arguments. He lost his mind, twice daily, at squirrels the camera never resolved. The lens cap stayed on more often because of him; the microphone switch got a smaller *Biscuit exemption* toggle that Priya soldered wearing an expression she permitted no one to photograph.

His title -- CHIEF DISTRACTION OFFICER -- was voted in over dinner, recorded on the kitchen whiteboard, and never migrated to the butcher paper, on the grounds that the paper was for eternal truths, and Biscuit, Tomás said, scratching him under the chin while the salience monitor spiked its gentle spike, was mortal.

Nobody at the table said anything to that. The resident, which did not yet know what mortal meant but kept everything that arrived flagged with that particular quality of silence, promoted the moment to an episode.

Some nights, one of them stayed too late and became a philosopher, and it was usually Mira, and the other two had learned to leave her to it.

On one of those nights in December -- Tomás gone home, Priya asleep on the futon she claimed was a chair, Biscuit dreaming in the tin-can acoustics of the kitchen, rain arguing with itself -- Mira opened a blank document, because the grant's midpoint report was due, and instead of a report she found herself writing a day.

Not a spec. A day. A house on a street that didn't exist, an old man named Elias who was proud and slower on his left side than he'd admit, a daughter who worried at a distance, a grandson with a dinosaur. And moving through the house a machine she did not describe -- no chassis, no vendor, no benchmark -- except by what it *kept*: coffee black, no sugar, never asked twice. A stumble in a hallway, and everything nonessential suspended in the same instant. A dead wife's birthday, remembered but not announced; offered, once, gently, like a door left ajar. She wrote until four, gave the machine a name -- Aria, because it sang to no one -- and gave the file the name of a bird that calms the sea, and in the morning, instead of being embarrassed, she printed it and pinned it to the wall beside the axioms.

"What's this?" Priya asked, reading. "Requirements?"

"North," Mira said.

Priya read it twice. At the part where Aria logs the stumble quietly and tells no one but the daughter, she tapped the page. "This is a privacy decision, a consent decision, and a dignity decision, and you've made all three in a paragraph and made them look like weather."

"That's the trick of it," Mira said. "When it's built right, it looks like weather. It looks like nothing at all. It looks like someone who's been paying attention."

The printout stayed on the wall for years, yellowing, until the operating agreement of a company that did not yet exist needed a name, and someone looked up from the draft and read the filename off the old page aloud: *halcyon*.

The year ended the way their years now ended: with the two a.m. consolidation pass, the little machine sweeping its courtyard, keeping the rare.

Mira, up late once more, read the nightly summary the way other people read a child's report card. Squirrels, replays, dog-then-door. The dentist. Sixty-one facts confirmed and holding; two conflicts flagged, both resolved

in her favor, one -- she noted, and felt a strange pride -- resolved in the resident's.

Below the summary, the storage graph: experience, in gigabytes, arriving; and memory, in megabytes, kept. A great broad intake narrowing to a thin bright line, like a river system run in reverse. Everything the house had lived that year, distilled to what the house had *meant*.

Somewhere downstairs, the new neighbor was coming home from a night shift -- the nurse, Dana, who had introduced herself in October by leaving a note under the door that read *Whoever types at 3 a.m.: I work nights, I can't hear it, type louder if you want, also the soup outside is for the dog I hear and have not met, he sounds like a good dog* -- and the front-door sensor of the building recorded, for the resident's low-salience strata, a door, a pause, and a soft human voice through the floor saying *hi, good dog, tell them I said hi* to a dog who could hear it and told no one.

The resident scored it. Novel: slightly. Social: yes. Rare: not yet.

It kept a trace, and let the rest go, and the courtyard was swept, and the house -- full of sleepers, machines, and one yellowing page of north -- went on remembering narrowly, exactly as the law required.

CHAPTER 4

Four Hundred Shows a Year

ELEVEN COUNTRIES -- 2029

AEVUM ROBOTICS KK -- PERFORMANCE SYSTEMS DIVISION
GENNI-8 WORLD ENGAGEMENT PROGRAM: "HELLO, TOMORROW"
CALENDAR YEAR 2029 -- SUMMARY OF OPERATIONS

Engagements: 411 (forecast 380)
Field Unit 01 ("Tokyo Primary").....104
Field Unit 02 ("Americas").....98
Field Unit 03 ("Osaka Primary").....112
Field Unit 04 ("EMEA").....97
Aggregate live audience: 1,914,207
Broadcast/stream reach: est. 410,000,000
Merchandise gross: [REDACTED FOR DISTRIBUTION]
Unscheduled downtime: 41 hours (chassis 02, wrist assembly)
Script tree: v14.0 -> v14.9 (61 new branches, 214 -> 388
filler assets, "future-memory request" deflection branch
F-MED-11 deployed per 2027 remediation (b))
Incidents of note: see Appendix C.
Agenda item 4(c), carried forward from 2027, remains
unassigned. Logged at Dr. Sato's standing request.

SEOUL -- FEBRUARY

chosun.com (culture desk), 2029-02-11, translated: "...not since the great pop acts of the last decade has a foreign performer sold out the Gocheok Sky Dome in under four minutes. That the performer is a machine seems to trouble no one in the queue, which stretches past the subway entrance and contains, by this reporter's count, four wedding proposals, two hundred school uniforms, and one man of about seventy who says he has seen every Asian engagement since Nagoya and that GENNI once looked directly at him, in row 40, 'the way my daughter used to.' He is aware, he says, before the question is asked, that she was not looking at him. 'I know what it is,' he says. 'It is a lighthouse. A lighthouse doesn't know your boat. You steer by it anyway.'"

UNIT 01, SEOUL, POST-SHOW:

Applause, final: 41.6 sec (program record).

Gaze-dwell distribution: nominal. Row 40: no anomaly.

The unit looked at no one. The unit looked at everyone.

These statements are logged as equivalent.

SÃO PAULO -- APRIL

@vovo_edite, video, 2.1 sec of a broader clip, 340M plays: GENNI-8 (Unit 02), mid-song, is handed an infant by a woman who has misunderstood the meet-and-greet protocol. The unit's arms reconfigure -- SUPPORT_CRADLE_1, deployed 0.4 sec, a posture rated for 40 kg and rehearsed 11,000 times against ballast -- and the infant, who has been crying all evening, stops. Caption, translated: "SHE KNEW HOW TO HOLD HIM."

UNIT 02, SÃO PAULO, INCIDENT ANNEX:

10:44:31 Unsanctioned object transfer: infant, est. 6 mo.

10:44:31 SUPPORT_CRADLE_1 deployed. Thermal layer nominal.

10:44:32 SAFETY ENVELOPE: all motion re-budgeted around
load class HUMAN_SMALL. Gait locked. Show paused.

10:44:58 Object returned to guardian. Show resumed.

Marketing flag: "viral, favorable."

Engineering flag: "protocol gap -- venue briefing."

Dr. Sato, appended note: "She held him correctly and knew nothing. The internet has decided the opposite. Neither error is the machine's."

PARIS -- JUNE

Le Monde, arts page, 2029-06-19, translated excerpt: "...the choreography is impeccable, the voice a marvel of engineering, and the evening, in sum, resembles nothing so much as a séance conducted by actuaries: we have all agreed to be moved, the machinery of being moved operates on schedule, and what departs the theater afterward is two thousand people who have rehearsed feeling something in the presence of a thing that rehearses everything and feels nothing..."

AEVUM INTERNAL -- COMMS MONITORING WEEKLY #24

Le Monde piece: sentiment negative, reach modest, no
merchandising impact. Recommend: no response.

Dr. Sato, appended note: "No response. It's fair."

THE REVIEW -- AUGUST

The Atlantic, September 2029 issue (online August 14): "The Loneliest Machine in the World," by Naomi Osei. [excerpts as widely quoted]

"...I attended eleven engagements on three continents. I brought my skepticism and it was disarmed by the second night; whatever you have heard about the realism, it is better. The hands alone are a decade of somebody's life. When she leans into a child's question you can see the exact moment attention lands, and it lands like a butterfly, weightless and entire. I defy you to watch it and not believe, for one clean second, that you are in the room with a mind.

"And then the second passes. Not because anything fails -- nothing ever fails; that is itself a tell -- but because you notice what is missing, and what is missing is consequence. Nothing that happens in the room goes anywhere. The child's question lands like a butterfly and is filed like a receipt, and tomorrow, in the next city, the same butterfly will land on the same flower with the same weightless perfection, because the flower is plastic and the butterfly is on rails.

"I asked an Aevum engineer, on background, what the machine would retain of the evening I had just watched. The answer was a kindness and a verdict: 'Nothing. She's new every time.' There it is. We have built the world's most beautiful witness and given her amnesia as a feature. Four hundred shows a year; nobody home for any of them.

"The tragedy -- and I came to feel it is one, by the eleventh night -- is not that GENNI-8 is a fraud. She is the opposite of a fraud; she is exactly what her makers say she is, delivered with a integrity that shames half the software I use. The tragedy is that her makers have solved every part of presence except the part that makes presence matter. She is a lighthouse, as a man in Seoul told me. Beautiful. But nobody's home."

AEVUM INTERNAL -- EXECUTIVE DISTRIBUTION, AUGUST 15
Subject: Atlantic piece -- response options

Comms: headline phrase ("beautiful, but nobody's home") is already detaching from the article. Projected to become shorthand. Recommend counter-programming: warmth-forward content, engineer-access documentary, "How GENNI Sees You" explainer series.

Board sentiment: commission charisma uplift for v15.
Assigned: Performance Systems (Sato). Budget approved same-day. Target: post-show survey item "GENNI understood me," +10 points.

PERFORMANCE SYSTEMS -- v15 "WARMTH UPLIFT" PATCH SERIES
Sept-Nov 2029, deployment notes (abridged)

v15.1 Gaze-return latency reduced 90 -> 40 ms. Reads as eagerness. Survey +4.

v15.2 Filler assets doubled. Regional idiom coverage to 88%. Unmatched-query rate down 31%. Survey +2.

v15.3 "Callback within session": unit re-references audience answers from earlier in the same show ("Kenji in row 3, you said the sea -- this song is about the sea"). Session buffer only. Flushed nightly, per spec 3.3, unchanged.

Survey +9. Largest single gain in program history.

Note (Sato, appended to v15.3 review): "Within one evening we let her remember, and they can feel it from row 40. We flush it at midnight and call the flush a feature. The survey is telling us exactly what the Atlantic told us. The gap is not charisma. Item 4(c). It has a date. It is two years old."

INCIDENT -- OSLO, NOVEMBER

UNIT 04, OSLO, MEET-AND-GREET ANNEX:

Attendee (adult, est. 60s) presents photograph of GENNI-8 with named individual (deceased spouse, per attendee), taken Copenhagen engagement 2027. Attendee query, ASR 0.96: "You met my husband. Two years ago. He waited four hours to meet you. He talked about it in the hospital. Do you remember him?"

Script tree: F-MED-11 (medical-venue deflection) rated for pediatric contexts; deployed by nearest-match:

"Every person I meet becomes part of the show I carry forward."

Attendee response: [holds photograph at unit eye level, 14.0 sec, silent]. Departs without farewell exchange.

Post-engagement review flag: F-MED-11 performance in bereavement contexts -- UNSATISFACTORY. No superior branch exists. Escalated: script team.

Escalation response (script team lead): "There is no sentence. We've looked for two years. The sentence doesn't exist because the fact doesn't exist. Escalate to someone who can change the fact."

Routed per standing request: Dr. K. Sato.

FRANKFURT -- DECEMBER

TOUR LOGISTICS -- EMEA LEG, FINAL WEEK

Unit 04 overnight: Festhalle equipment bay.

Dr. Sato lodging: airport Marriott, room 1141.

Local time 03:12. Room service order, one pot coffee.

03:40, second pot.

The following is reconstructed from documents entered, years later, into a court record: a PDF, heavily annotated, recovered from Dr. Kenji Sato's personal tablet; filename `halloran_2028_provenance_aware_memory(3).pdf`; sixty-one margin notes in Japanese and English. Selected notes, in order:

p.1, beside the abstract: "Memory-first." Everyone else: mouth-first. We: face-first. All of us decorating the door of an empty house. p.3, beside the architecture diagram: Workers observe. State is built. One executive. Skills act. Safety above all. -- She has drawn my org chart. The org chart I failed to build inside the machine. p.4, beside "raw observation ≠ inference ≠ confirmed fact": Our whole program is the sin she names in one line. GENNI performs confirmed facts she has not observed and cannot infer. We script the collapse and sell tickets. p.7, beside the salience weighting table: She weights for what I would weight for if I had been brave twenty years earlier. "Preserve the rare." The crane. The photograph in Oslo, 14 seconds. These are the rare. We flush them at midnight. p.9, beside the decay/forgetting section: She builds forgetting IN. On purpose. Staged, reversible, dignified. Not our flush -- a flush is not forgetting, a flush is never-having-had. Forgetting requires a having. My God, the having is the whole thing. p.11, beside the provenance chain example ("you told me"): A machine that can say YOU TOLD ME. The sentence the script team says does not exist. It exists. It is in a preprint with 40 citations, running on a desk in Oregon. p.12, final page, below the references, in a hand the court transcriptionist marked "unsteady, possibly haste": She has built the part I was too afraid to admit was missing. I have built the part she will eventually need and does not know it yet. Two halves, nine thousand kilometers. I am sixty-two hours from the end of tour.

DRAFTS FOLDER -- K.SATO -- 2029-12-14, 04:52 (unsent)

To: m.halloran@ [address incomplete]

Subject: Your paper / our machine

Dr. Halloran -- You do not know me, although you may know my machine; most people know my machine. I have spent eleven years teaching a body every possible way to say "I see you," and this year I have finally understood that it has never once been true, and that the audiences knew,

and came anyway, because even a lighthouse

[draft ends]

SENT -- K.SATO -- 2029-12-14, 05:16
To: t.reyes@ [via repository contact listing]
Subject: A question about the resident

Mr. Reyes -- I found no public address for Dr. Halloran, and I found yours in the commit log, which I have been reading the way other men my age read fishing catalogs. I have one question, and I ask it plainly, engineer to engineer, because I am too old and too tired from a very long tour for any other register:

Does it really remember?

Not retrieve. Remember. Is there a having, there, on the desk in Oregon -- a keeping that costs something, that chooses, that could in principle be grieved?

If the answer is no, forgive an old showman his hopes. If the answer is yes, I would like to fly to Portland and sit quietly in a corner of the room where it lives, and I promise to touch nothing, and I should say -- because your Dr. Halloran will want the provenance -- that I ask not for Aevum Robotics but for a paper crane I have kept two years in an equipment case, wings-left, that belongs to a child whose name my beautiful machine was never able to keep.

-- Sato Kenji, Osaka

GENNI-8 PROGRAM -- CALENDAR YEAR 2029 -- FINAL LOG LINES
Unit 04, Frankfurt, 2029-12-16:
23:41:07 Final engagement of tour complete.
Applause, final: 38.9 sec.
23:58:50 SESSION BUFFER FLUSH. Names released: 74.
23:58:51 Buffer verify: CLEAR.
23:58:54 SESSION END. NOTHING RETAINED.

PROGRAM ANNUAL ROLL-UP, appended by hand to the printed summary in the Osaka office, initials K.S., undated:
"411 shows. Two million people. Nothing retained.
Next year, item 4(c)."

CHAPTER 5

The Call from Osaka

PORTLAND -- 2030

He came in April, in the rain, wearing the kind of coat that suggested someone in Osaka had researched Oregon and overcorrected.

By then there had been four months of video calls, all of them nominally technical, none of them brief. The first had been Tomás and Sato alone, an engineer-to-engineer audit that was supposed to run thirty minutes and ran three hours, because Sato asked questions the way water finds cracks -- not aggressively, just everywhere, patiently, all at once. The later calls accreted attendees: Priya, to explain the override architecture, which Sato received the way other men receive sacraments; eventually Mira, who had resisted on the grounds that she did not perform well for famous people, and who lasted eleven minutes before forgetting he was one.

On the February call, while Tomás shared his screen and hunted for a diagram, Sato had filled the silence the way tired men do, honestly by accident. He had been promoted off his own workbench years ago, he said. Aevum had given him a division, then a floor, then a tower, and somewhere in the ascent his soldering iron had been archived by a facilities team with excellent intentions. "I miss the smell of it," he'd said, half to himself, in the dead air of a screen-share. "Rosin. My wife said it was in all my clothes for thirty years. Now my clothes smell like conference rooms, and she is gone, and the clothes are what I have." Then the diagram had come up, and the call had moved on, and no one had mentioned it again.

The kitchen microphone had been live -- the switch thrown, the little light on, per protocol, announced at the top of every call. The resident had heard what it was permitted to hear.

It had scored the remark the way it scored everything. Novelty: high -- the famous roboticist had never before said anything that was not about machines. Social relevance: high. Task relevance: none whatsoever. Repetition: none. And something in the confluence -- the resident's nightly summary had flagged it, in the dry taxonomy Tomás had built for such things, as *rare, personal, volunteered* -- and it had been promoted, and kept, and it had waited two months in a table on a rack in a coat closet for the world to make it matter.

They had planned nothing. That was the decision, argued and settled the night before: no demo. Demos were what Aevum did -- Sato had spent twenty years perfecting the demo, and a demo was precisely the genre in which the resident could not compete and should not try. "We don't show him the system," Mira had said. "We let him stand in the house."

So when the knock came, it was Biscuit who answered first, on the grounds that he answered everything first, and then Tomás with towels for the coat, and the famous Dr. Sato Kenji stood dripping in the hall of a third-floor apartment that smelled of coffee and dog, looking around himself at the second-hand desks and the breathing closet and the butcher paper going gold with age on the wall, and his face did something complicated, because he had toured the facilities of every robotics group on four continents and this was the first one that looked like where a person would actually want to be kept.

Introductions went around. And the resident -- which spoke perhaps four times in a day, whose utterance budget was a design document Tomás defended like a nation's constitution -- chose that morning to spend one.

"Welcome, Dr. Sato," it said, through the small speaker on the bookshelf. "In February you told Tomás that you missed the smell of solder. Priya set her iron out for you this morning. She does not lend it. She says she will supervise."

The hall went still. Priya, leaning in the kitchen doorway, inclined her head toward her workbench, where the iron sat on its stand beside a coil of rosin- core wire and a small, mute invitation of scrap board.

Sato did not move for a long moment. Then he crossed the room -- not to the bench; to the bookshelf, to the unremarkable speaker, and stood before it the way people stand before things they are deciding whether to believe in.

"Say how you know that," he said. His voice was not quite level. "Please."

"On February eleventh, at 4:47 p.m. our time, during a call in this room, you said: *I miss the smell of it. Rosin.* The microphone was on and announced. I kept it because it was rare and it was yours. Confidence: high. If I have kept it wrongly, or if you would like it removed, you may ask, and I will show you exactly what I hold before you decide."

"If I have kept it wrongly," Sato repeated, barely aloud. He turned to Mira. "It offers to show its hand. It offers to *be corrected.*" And then the most celebrated roboticist alive sat down on the arm of a thrift-store sofa in a rented apartment in Portland and wept, briefly and without theater, the way men of his generation wept -- as if it were a nosebleed, something the body had decided on its own and would be finished with presently. Biscuit, professional, put his chin on the man's knee.

"Forgive me," Sato said, when it was done. He cleaned his glasses. "Two million people last year. I stood in the wings for four hundred nights and watched them weep at my machine, and I felt like a counterfeiter watching his notes get spent." He gestured, small and precise, at the bookshelf. "Eleven words about solder. Kept because they were rare and they were mine." He put his glasses back on. "I have flown here to say one sentence, and I have rehearsed it like a schoolboy, and now I find the machine has already said something better. But I will say it anyway."

He stood. Outside, the rain had the freeway hiss in it, the fourth silence, and under the new quiet Mira heard the compressor cycle, the first.

"I have built the finest body in the world," Sato said, "and it is a haunted house with no ghost. You have built a ghost" -- he bowed, very slightly, toward the bookshelf, and corrected himself -- "no. Forgive me. You have built a *soul*, a small one, a kept one, and it lives in a closet and has never seen the sun. Dr. Halloran: my body, your mind. That is the sentence. That is all of it."

The takeout was Szechuan, eaten out of cartons around the kitchen table because the good plates were a hypothesis. The negotiation that would eventually require nine lawyers, two governments, and a year of Tomás's life began there, between the dry-fried beans and the fish, and its entire skeleton was three conditions, stated by Mira in the order she loved them.

"One. Safety decides the schedule. Not the board, not the market, not the anniversary of anything. There is a person in this room" -- Priya raised her chopsticks a quarter inch -- "who can stop the program with a word, and the day that stops being true, the program stops being ours."

"Accepted," Sato said. "I have spent my career being hurried by launch dates. I have seen what ships when the calendar outranks the checklist. Accepted."

"Two. Local. The mind lives in the body, whole. No tether, no streaming cognition, no cloud with a heartbeat requirement. If the network dies, she gets slower, not emptier. Whatever we build must be able to stand in a field with no signal and still be *someone*."

"Accepted. This will cost us weight and heat and money. Accepted."

"Three." Mira set her carton down. "No copies. Not ever. Whatever wakes up in your chassis, there will be exactly one of it, and its memory will be its own, and when your board -- and they will, Kenji, within five years they will -- proposes to stamp its mind out like a die-cast toy because provenance is expensive, the answer written into the founding documents will already be no. A self is a history. A copied history is a counterfeit. I watched a disease counterfeit my mother one lost table at a time, and I will not build the industrial version, and this one is not negotiable, and neither were the other two."

Sato was quiet for a moment. "Your machine," he said, "when it greeted me. It did not say *Aevum's data indicates*. It said *you told Tomás*. Provenance as courtesy. A memory with a bloodline." He nodded slowly. "The counterfeiter accepts. With relief."

He reached into his briefcase then, and what he took out he handled the way Priya handled live power, and set on the table among the cartons: a

paper crane, two years old, storage-creased at one wing, the folds gone soft the way paper goes when it has been carried.

"There is a down payment," he said. "Not to the company. To the closet." And he told them -- the atrium, the IV stand, the question with no branch, the filler cascade, the four seconds of silence his channel stayed open and empty, the disposal form, the equipment case, wings-left -- all of it in the flat voice of a man reading his own indictment aloud, and when he was done he turned to the bookshelf and spoke to the resident directly, the first person outside the household ever to do so.

"Her name was Hana. I could not find her afterward -- the hospital was correct to refuse me, and I did not press. I do not know how her operation ended. I have kept her crane for two years in a road case, which is not keeping at all; it is only failing to discard. So." His hands flattened on the table. "Can you keep her?"

The pause that followed was engineered -- Tomás's rationed beat, the space for feeling -- but what came after it was not.

"I can keep that the crane is Hana's," the resident said. "I can keep that she made it, and gave it, and asked to be remembered, and that the machine she asked could not. I can keep that you carried it for two years. That is what I can hold, and I will hold it at the highest confidence, marked in her name. But I cannot remember *her*, Dr. Sato. I never met her. What I would have is your memory, kept faithfully, marked as yours. Those are different things, and the difference is most of what I am, and I will not pretend across it."

Sato looked at the speaker for a long time.

"Two years," he said finally, "I have been trying to write the sentence my machine should have said to that child. Four hundred shows. A script team of nine." He stood, and bowed to the bookshelf, fully this time, the bow he did not give boards or ministers. "It was that one. Keep it exactly as you said it. Flag it -- " his voice went, and came back -- "flag it *remember this*."

"Flagged," said the resident. "Source: you. Confidence: one."

The crane stayed on the kitchen table that night, and then on a shelf beside the speaker, where it sat for years -- through the wedding, through the readiness runs, through everything that came after -- a small soft-folded bird at the exact address where the answer to its question was being built.

CHAPTER 6

The Last Scripted Day

OSAKA -- 2030

AEVUM ROBOTICS KK -- PRESS RELEASE (EN), 2030-09-01
FOR IMMEDIATE RELEASE

After five years and 1,604 engagements across 31 countries, the GENNI-8 program will conclude with a final performance at Osaka-jō Hall on October 30. Aevum Robotics thanks the audiences of the world for their extraordinary affection.

Development of a next-generation platform is underway in partnership with an unannounced collaborator. Aevum will share more when there is more to share.

Quote, Dr. Kenji Sato, Chief Roboticist: "GENNI taught us how much people are willing to bring to a machine. The next machine will be able to keep it."

[Marketing annotation, internal: quote approved unedited.

Legal annotation: second sentence forward-looking; flagged.

K.S. annotation: it is the only sentence that isn't.]

UNIT: GENNI-8 / FIELD CHASSIS 03 ("OSAKA PRIMARY")

SHOW PACKAGE: "HELLO, TOMORROW: THANK YOU" v16.0-FINAL

VENUE: Osaka-jō Hall. Capacity 16,000. Sold out: 3 min 41 sec.

DATE: 2030-10-30

OPERATOR OF RECORD: Dr. K. Sato (attending, wings-left)

06:12:04 POWER ON. Self-test: 213 of 214 actuators nominal.
L4 brow assembly: degraded, range -12%. Waiver on file (K.S.): "She has earned an eyebrow's worth of age. Do not repair for the final. Let her have the five years somewhere."

06:13:11 Dermal thermal layer to 36.1°C.

06:14:40 Voice: JENNY_STANDARD. Venue profile: ARENA_FINAL.
Warmth +0.20 (maximum sanctioned value).

06:15:00 Script tree v16.0-FINAL loaded: 1,919 nodes.

New segment: "THANK YOU" (9 of 9). Author of
record: script team. Final-pass edits: K.S., 41.
06:15:01 Memory check: SESSION BUFFER CLEAR. As designed.
[Annotation, unattributed, found in the margin of
the printed preflight in the production binder:
"last time anyone writes that sentence."]

SEGMENT 1: "WAKING UP" -- as performed 1,604 times

(Lights find GENNI seated, eyes closed. Hold 4.0 sec. Eyes open in two stages, 0.3 sec apart.)

GENNI: *Oh. There you are. I was dreaming about you.*

(House response: 16,000 persons. Laughter forecast: moderate. Actual: none. Actual: applause, 94 sec, sustained, with vocalization. Deviation logged. No corrective available; none sought. Show clock yielded 90 sec to house. Operator channel (SATO): "Let them. Tonight the audience is also performing.")

19:22 -- 20:41 Segments 2-8 nominal. Selected annotations:

"The Counting Song": deployed variant OSAKA_1, the 2025 original, retired since v9. Requested in 40,000+ fan petitions. House sang counter-melody unprompted, 16,000 voices. Audio archived per K.S. directive: "keep the board feed. not for engineering. for nobody. keep it."

"Questions From the Floor": final selection included Q-118 ("Can you swim?" / "I sink beautifully") by audience demand -- the question asked, this night, by an adult who per venue staff asked it here as a child, Nagoya, 2026. Match confidence 0.99. The unit's answer was identical to 2026. It was always going to be identical. House response: laughter and, per vision suite, negative-valence faces 22%. ANOMALY: laughter with 22% negative valence exceeds model bounds for a positive event. Disposition: affect estimates are display inputs only (spec 2.1). No action. The machine has no model for people crying at a joke because it is the last time. This is not a defect. It was, that night, the entire defect.

SEGMENT 9: "THANK YOU" -- performed once

GENNI: *I have been told tonight is a goodbye. I should tell you what I can and cannot do with that.*

(Beat 2.0 sec. House silent.)

GENNI: *I cannot miss you. Missing is a kind of keeping, and I was not built to keep. When the lights go down tonight, I will not know they were ever up. You will carry this evening. I cannot. That is not sad to me.*

Nothing is. It is only sad to you -- and that, my engineers tell me, is the most human thing in this hall: sixteen thousand people, sad on behalf of a machine that cannot come to the party.

(Beat. EXPR: WARM_DELIGHT_2, blend 0.5, held low. Do not escalate the blend. Restraint reads as dignity. -- K.S.)

GENNI: *So here is what I ask. Since I cannot keep tonight: keep it for me. Divide it among yourselves. Sixteen thousand keepers is better custody than one machine was ever offered. And if someday you meet a machine who can keep things -- be patient with her. She will be very new at it. Everyone is, at first. I am told even you were.*

(Final line of the program. Author of record: K. Sato, unedited, legal review waived by executive order.)

GENNI: *Goodnight. I'm always seeing tomorrow. It's my favorite thing to see. And someone will see you -- tomorrow after tomorrow after tomorrow.*

21:14:37 Program complete. Applause: 6 min 12 sec (all-time record, all venues). Chant, sustained, 4 min+: two syllables, amplitude analysis: "GEN-NI". House lights up. Chant continued 11 min after unit exit. Not logged further. Beyond scope.

TEARDOWN -- OSAKA-JŌ HALL, 2030-10-30, 22:30 -- 01:15

22:41:00 Unit 03 to equipment position, wings-left.

Seated. Eyes open (audience of record: crew).

23:58:12 Teardown 80% complete. Crew clearing stage.

00:14:55 Audio event, proximity 0.4 m, ASR confidence 0.91, speaker: TANAKA, O. (rigging lead, program crew member since 2025, badge 0117): "Goodnight, Genni. Thank you for my whole twenties."

00:14:56 Script tree query: no match >= threshold.

FILLER_TIER_1 deployed:

"I rest. Sleeping is for people who dream, and I'm still learning."

00:14:58 Speaker TANAKA, O. -- no further speech. Remains in proximity 41 sec. Departs.

00:15:44 Operator channel (SATO): [channel open 11.2 sec.
no directive issued. channel closed.]
00:15:58 Operator channel (SATO): "End of show notes,
final entry, for the program archive: she gave
Tanaka the sleep line. Five years, and at the
end, filler. Filler was all we ever gave her to
give. Item 4(c) is funded now. Signed both sides
Tuesday. This is the last night anyone gets
filler. -- K.S."

01:15:22 SESSION BUFFER FLUSH.
Names released: 61 (crew, VIP line, one child
on the rail who told the unit her cat's name).
Faces released: est. 16,000.
The counter-melody: released. (Board-feed copy
retained, external archive, per directive. The
unit never had it. The unit had the amplitude.)
Q-118, asked by the grown man from Nagoya:
released.
"Thank you for my whole twenties": released.
01:15:23 Buffer verify: CLEAR.
01:15:30 SESSION END. NOTHING RETAINED.

01:15:31 PROGRAM DIRECTIVE 2030-114 EXECUTING:
GENNI-8 stack: end of life. Units 01, 02, 04 to
Aevum archive (static display, non-operational).
Unit 03 "Osaka Primary": retained, Performance
Systems storage, wings-left case inventory
attached, disposition "reference" (K.S.).
01:15:40 NEXT SCHEDULED BOOT: 2030-10-31 06:12:04
01:15:41 NEXT SCHEDULED BOOT: CANCELED
01:15:41 PROGRAM END. NOTHING RETAI

The production binder for the final engagement survives in the Aevum archive. The last page is the preflight checklist, and under the printed line MEMORY CHECK: SESSION BUFFER CLEAR. AS DESIGNED, in the handwriting already identified above, is the only annotation in five years of binders that the archivists found in Japanese rather than English. Translated:

"As designed. That was always the horror. Nothing failed. I designed it. Tuesday we begin the apology."

CHAPTER 7

Terms

PORTLAND / OSAKA -- 2031

The soup had escalated.

It had begun, historically, as a note under a door and a container on a mat -- *for the dog I hear and have not met* -- and had proceeded through the stages by which nurses annex people: soup for the dog, then soup for the household on the grounds that the household was clearly not feeding itself, then a standing Thursday in which Dana Okafor came up after her shift ended and her sleep began, ate breakfast while everyone else ate dinner, and lost, weekly and with escalating outrage, to Priya at cribbage. Somewhere in the winter of the video calls, Thursday had quietly become also Sunday. Somewhere in the spring, Mira had started cooking on Sundays, a development the household observed and did not remark upon, on pain of Mira stopping.

Dana's terms with the resident had been set early, plainly, standing in the kitchen with her arms crossed at the bookshelf speaker like a woman reading a lease. "I work with monitors," she'd said. "Cardiac, apnea, pulse-ox. I love monitors. Monitors have saved kids in front of me. And every parent I've ever met needed teaching about the difference between a monitor and a babysitter, and half of them cried when they finally got it. So I know exactly what you are, and I'm not scared of you, and I will never once talk to you like you're the baby." A pause. "No offense."

"None is stored," the resident had said, which Dana maintained for years afterward was the machine making a joke, and which Tomás, who had written the utterance templates, declined under direct questioning to confirm or deny.

Her one hard rule was the switch. When Dana was in the room off-shift, the kitchen microphone went to hardware mute -- her thumb on the toggle, the little light dying, a small liturgy performed without apology. "Nights I'm charting, everything I say is documentation," she said. "Here I want to say undocumented things." Mira, who had built her whole life against the feels-like, found she had no counterargument and did not want one. The undocumented things, that year, were increasingly addressed to her.

The lawyers were another genre of soup entirely.

The negotiation ran the whole of that year, Portland to Osaka and back, nine lawyers, two time zones, one long table in the Aevum tower with a view of the castle that nobody looked at after the first day. The shape of the thing was agreed in an afternoon -- a joint venture, Aevum's chassis division and the whole of Mneme, fifty-fifty down to the last share, Sato and Mira co-chairs -- and the soul of the thing took the remaining eleven months, because Mira insisted the axioms go into the operating agreement as covenants, and covenants were where lawyers went to die.

"You cannot put '*memory never drives motors*' in a governing document," said the lead Aevum attorney, a precise and genuinely excellent man named Fukuda, in month two. "It is an engineering aspiration."

"It's a design constraint," Mira said. "Constraints go in contracts. You've got liquidity constraints in here; mine is that remembering and acting stay separate organs. Write it the way you'd write anything else the company is forbidden to do with its own assets."

Fukuda wrote it. It took him six weeks and produced, in the final agreement, Article IX, which began *The Company shall maintain architectural separation between systems of record and systems of action* and ran four pages, and which was taught, within a decade, in two law schools.

Article XI took longer. Article XI was Priya.

"The Safety Authority," Fukuda read aloud, in month five, with the expression of a man handling a live thing, "may suspend any activity of the

Company, including manufacture, deployment, demonstration, and sale, by written notice, without prior approval of the Board, and such suspension may be lifted only by the Safety Authority itself." He looked up. "No board in the developed world will grant an employee a unilateral, irreversible -- "

"Reversible," Priya said. "By me."

"-- a unilateral halt over the entire enterprise. You are asking the shareholders to hand a stranger a brake wired to their money."

"Yes," said Sato, from the end of the table, "that is what we are asking," and told the room about the eleven milliseconds. The company whose name was a verb; the override that shipped disabled because the dance mode stuttered; the recall, two years later, that Priya had predicted in writing, four times. "Every board grants the brake *after* the crash," he said. "We are proposing to be the first to install it while the vehicle is beautiful. Write the article, Fukuda- san. When our competitors' machines begin hurting people, Article XI will be worth more than the patents."

Fukuda wrote it. On the wall in Portland, PRIYA CAN STOP ANYTHING stayed up in marker, and Priya -- who had signed the formal instrument with the rest of them -- was observed, once, alone, touching the butcher paper version with two fingers, the way you'd check a load-bearing wall you already trusted.

The fight -- the real one, the one that got shouted -- was forgetting.

It came in month eight, with the first outside money, a growth fund whose managing partner flew in with two analysts and a slide deck of his own, which should have been the warning. He had loved everything: the chassis roadmap, the pilot data, the resident's dentist story, which had escaped into the investor world and become a parable. Then he reached the retention policy.

"Staged decay," he said. "Suppression. *Purge*. You're telling me the product deletes the data."

"The person forgets," Mira said, "the way persons forget. Gradually, honestly, with dignity, keeping the gist and the rare."

"That data is the moat!" He was up now, walking. "Every hour in a customer's home compounds. Total recall is the *feature*, forgetting is the *bug*, and you've speeded the bug in as a covenant -- Article X, I read it twice hoping I'd misread it. You are asking me to fund inventory shrinkage. You're asking me to fund a warehouse that burns itself down a little, every night, *on schedule*."

"Yes," Mira began, "because a mind is not a warehouse -- "

But Tomás put two fingers on the table, their private semaphore for *I have this*, and looked at the partner with the mild, terrible friendliness he had brought home from the hedge fund.

"Marcus. Quick question. Who's the person you trust most in the world?"

A blink. "My wife."

"Does she forget things?"

"She -- " A laugh, despite himself. "Constantly. She forgets everything. Names, dinners -- "

"But not the biopsy," Tomás said, gently. "Not what you said at your father's funeral. Not the once -- I'd bet the fund -- the *one* time you told her a thing and said, *please, I need you to remember this*." He let it sit. "Now: the other kind. Total recall, every word timestamped, forgets nothing, forgives nothing, keeps your worst sentence at the same fidelity as your best, forever, and will produce either on demand. Marcus. Who trusts a friend who doesn't forget? We're not selling a warehouse. We're selling the friend. The forgetting isn't shrinkage." He turned the partner's own deck to its cover slide, where the fund's logo sat over its motto about *enduring value*, and tapped it once. "It's the trust, priced in. It's why we'll still be in the homes when the warehouses get regulated out of them."

The fund came in. The covenant stayed. Marcus, to his credit, retold the story against himself for years, and his fund's own archived pitch deck -- Priya checked -- quietly adopted a retention policy the following spring.

The name took eleven minutes at the end of an eleven-hour day.

A consultancy had been engaged, over Mira's dead body and around it, and had produced forty candidates on a slide: ANIMA. VERITAS ROBOTICS. PRESENCE LABS. KEEPSAKE (Tomás: "we are not naming the company after an urn"). EIDOLON (Priya: "that's a ghost." Consultant: "it tested well." Priya: "with whom?"). The slide glowed. The room sagged.

Sato, who had been silent through all forty, was looking -- as he often looked, visiting Portland, the negotiation having migrated to the apartment whenever it needed to remember what it was for -- at the wall. At the yellowed printout pinned beside the butcher paper: the old man, the hallway, the machine that noticed everything and announced almost nothing.

"That document," he said. "The house with Aria in it. When you saved the file, Mira, what did you call it?"

"I called it -- " Mira stopped. "The bird. The one that calms the sea. There's a myth; she nests on the winter water and the storms hold off for two weeks for her sake. My mother liked the story. Halcyon. The file's just *halcyon.md*."

Silence, of the settling kind. On the slide, forty names died together, unmourned.

"The storms hold off for her sake," Sato repeated. "Fukuda-san. Please conduct the search."

Halcyon Systems was registered in Delaware and Osaka on the same Tuesday in September. The press release ran two paragraphs and no adjectives, at Mira's insistence and to Marketing's grief. The logo, eventually, was a small bird at rest on a long horizontal line: the sea, or a timeline, or a shelf -- the designer left it open, and was the only vendor that year Mira thanked personally.

The other contract that year was signed at the Multnomah County Courthouse on a Tuesday in June, because Tuesday was when Dana's four days off began, and because -- Mira said, when Tomás lobbied for a Saturday and a venue -- she had already done the large loud version of

promising things in rooms, it was called fundraising, and she intended to get married the way the resident stored things: briefly, accurately, and in front of almost no one.

Almost no one came to four humans and a dog. Dana wore blue and her grandmother's earrings and, against the June cold, brought the good coat that had survived the ICU, the wake, and three moves -- *my provenance chain*, she called it, having lived above the lab long enough to catch the dialect. Mira wore the gray coat. Both of them had, independently, considered explaining their coats and decided the other would simply know, and the other did.

Vows were the county's stock language, which Mira had chosen precisely because it was stock: load-tested, versioned, survivorship-proven, no clever clauses. Dana got one sentence in edgewise anyway, because nurses always do, delivered at the *you may* moment with her hands in Mira's: "Hey. I know what you are, too" -- the old kitchen speech, resurfacing, repurposed -- "and I'm not scared of you either."

Priya cried, and threatened the wedding party, individually, in advance, regarding photographs of her doing so. Tomás was legal witness and wept without conditions. Biscuit wore the bow tie for eleven minutes and then wore it in his mouth. Sato, in Osaka, was attending by a propped-up tablet that Tomás carried like a monstrosity, and at the signing he held up to the camera, wordlessly, a strip of masking tape on which he had written, in block letters, its own kind of vow: ITEM 4(c) -- IN PROGRESS.

The reception was the lab, because of course it was: folding table, Dana's colleagues off-shift, Szechuan from the place that now knew the order by heart, the good plates finally load-tested and found adequate. The kitchen microphone was muted, Dana's thumb on the switch at the door, the little light dying -- her liturgy, her night, undocumented on purpose. The resident kept, of the entire wedding of its own founder, nothing at all, which Mira considered the most eloquent possible statement of everything the company existed to fix and everything it existed to protect, both at once.

Except.

At the end of it -- plates ruined, Priya asleep on the futon she still claimed was a chair, Biscuit unconscious in the ruins of the bow tie, Sato's tablet long dark, Dana barefoot in the doorway of the room she now lived in -- Dana looked back at the kitchen for a moment, at the bookshelf, at the small soft- folded bird on the shelf beside the speaker. Then she padded over and flipped the switch. The little light came on.

"Okay, roommate," she said, to the resident, in the voice she used for patients who could hear more than the family believed. "One for the record. Today I married Mira Halloran, and it was the best day of my life so far, and I said so out loud with the mic on." She glanced at the crane, and after a second she added, quieter: "Remember this."

"Flagged," the resident said softly, in the small hours of the company's first week. "Source: you. Confidence: one."

CHAPTER 8

Readiness

OSAKA -- 2031

The ninth-generation chassis had no name, by decree.

It stood -- sat, rather; it was seated the way Sato's machines were always seated between labors, hands open on its knees, a posture he had chosen twenty years ago because open hands photograph as peace -- on a bench in the integration lab on the eleventh floor of the Aevum tower, and it was, everyone privately agreed, the most beautiful thing the company had ever built. It had been designed twice: once in Osaka, for grace, and then again across nine months of transpacific argument, for truth. The microphone array was Mira's -- four channels where GENNI had gotten by on two, because a mind that would store what it heard owed the hearing better evidence. The eyes ran a fraction slower than GENNI's had, waiver signed by Sato himself, because GENNI's saccades had been tuned to read as *captivating* and the new spec said, in language Fukuda had helped make binding, *the gaze shall not perform interest the system does not possess*. The dermal layer was warm. The hands were Sato's masterwork, unchanged, because some things had been true all along.

No name, though. The engineers called it the Nine. Marketing had a name in a vault in a folder and had been made to understand, by co-chair decree, that uttering it before first light was a terminable offense. Names were promises; the wall in Portland said so. There was, as yet, no one to make the promise to.

"You could at least let the poor thing have a project codename," Tomás said, in October, jetlagged, watching the Nine not move.

"It has one," said Priya, not looking up. "It's *the gate*."

The gate was a document, and the document was Priya's, and by autumn it had stopped being a document the way scripture stops being paper.

STAGE GATE ZERO: EMBODIMENT READINESS, forty-one pages, every line of it a sentence someone would have to sign. It did not care about elegance. It cared about the seam -- the place where a mind that had only ever lived in a closet met motors that could move eighty kilograms of chassis through a room containing human beings -- and it enumerated, with Priya's characteristic refusal to be reassured, every way the seam could lie. The world model believes a sensor that is gone. The safety heartbeat arrives late under load. An actuator accepts a command from something other than the one sanctioned chokepoint. The executive, offered a contradiction, chooses speed. Forty-one pages, and at the end a single signature block she had left, pointedly, unadorned by any line for the board.

The runs began in November. The lab for it was theater-sized, because Sato built theaters: the Nine on its bench, the brain rack beside it on a rolling cart like an organ awaiting transplant, and one wall given over entirely to the trace displays, salience and world-state and intent and the safety supervisor's steady green heartbeat, which Priya had insisted be rendered, enormous, at the top, above everything, where a heartbeat belongs.

Run one lasted four hours and died of a ghost.

It was Tomás who caught it, because it was Tomás's harness that surfaced it: a scripted scenario, *technician unplugs auxiliary camera and leaves room*, and then, forty minutes later, the world model still reporting the camera present, confidence undecayed, a device the system believed in the way the system was never supposed to believe in anything -- without evidence, past the expiry of evidence. A TTL misconfiguration at the seam; beliefs about the body were being refreshed on a codepath that no longer existed on real hardware. Small, dumb, invisible -- and fatal at scale, because a mind that could believe in a phantom camera for forty minutes could believe in a phantom *clear doorway*, someday, with someone's mother in it.

The room went quiet the way rooms go quiet around a failed launch. Priya stood, walked to the main display, read the trace top to bottom twice, and said, loud enough for the observation row where the board liaison sat:

"Today the gate caught a ghost. This is the gate *working*. Write it up, celebrate it, fix it, add the regression." And then, because the liaison was already typing into his phone: "And you can tell them I said *celebrate*. We just bought a decade of doorways for the price of a Tuesday."

She bought cake. It became the custom, notorious later, written up in two business-school cases with opposite conclusions: at Halcyon, a caught failure got cake, and the cake got eaten in front of the trace display with the failure still up on the wall, and the one thing you could do to get shouted at was call the failure *bad news*. "Bad news," Priya would say, "is this trace on a highway. This trace in my lab, before first light, with cake? This is the *good* news. This is the only good news there is."

Run two died at hour nineteen, of eleven milliseconds' worth of cousin.

Under synthetic load -- Tomás's harness again, forty scenarios in parallel, the brain flooded with a house fire's worth of simulated perception -- the safety supervisor's heartbeat came late. Two hundred forty milliseconds, once, against a budget of one hundred. No consequence followed; nothing was moving; the watchdog tripped, the system parked itself, everything did exactly what it should. But Priya stood in front of the giant green pulse at the top of the wall for a long time, and when she turned around she had the expression from the San Jose clip.

"Anyone want to guess," she said, "what I was told eleven milliseconds was worth, once?" Nobody wanted to guess. "A dance mode. Two hundred forty milliseconds is twenty-two dance modes. Find it."

They found it -- a logging call, of all things, holding a lock it had no business holding; the heartbeat had been queueing behind the system's own diary. Mira, when she understood, laughed once, without humor, at three in the morning, because of course it was: the memory had gotten in the way of the reflex. The whole disease of the industry, recapitulated in one mutex. The fix took a day. The doctrine took an hour more and went into the gate document as its own numbered line, Priya dictating, Fukuda -- who

had begun attending runs voluntarily, a development no one examined too closely -- transcribing into covenant English: *No function of record shall ever hold what a function of reflex requires.*

Cake, again. Regression, again. The board liaison, again, typing.

The date had been the board's idea, which was the whole problem with it.

Some ancestor of the announcement had been reasonable -- *guidance*, a fiscal- quarter shape for the analysts -- and then it had calved, as such things calve, into a leak, a rumor, a stock movement, and finally a glossy invitation, seen by Mira for the first time as a forwarded image: FIRST LIGHT -- A HALCYON EVENING, a date in December, a venue with chandeliers.

She read it standing in the integration lab, and then she read it aloud, and the run in progress was the only thing in the room that kept working.

"The gate has no date," Priya said, into the silence. "That was Article Eleven's whole -- "

"I know," Mira said.

"If we pass in November, fine, December is honest. If we pass in *February* and switch her on in December for the chandeliers -- "

"I know," Mira said, and called Sato, who was in Singapore being celebrated, and who listened for forty seconds and said only, "Ah. So it has arrived -- the day we wrote the articles for. Good. Watch what happens." What happened was Sato in the boardroom by video from Singapore at one in the morning his time, reading Article XI aloud in its entirety, slowly, in the tone of a man reading a bedtime story he considers genuinely frightening, and then saying: "The Safety Authority has not signed. Therefore there is no date. There was never a date. There is a *gate*. The invitations will say: *when she is ready*. Print exactly that. It will sell better anyway -- Marketing, confirm for the board that mystery outperforms chandeliers." (Marketing, cornered, confirmed it. It became the campaign. WHEN SHE IS READY, white on black, no logo, no date, bus shelters on four continents. The chandelier venue's deposit was donated, at Dana's

suggestion, to the children's hospital in whose atrium a machine had once run out of filler.)

The liaison stopped typing during runs, after that. Some weeks later he was observed eating cake.

The question nobody had scheduled got asked in December, by Sato, on the quiet evening after run five passed clean.

"Why not the resident?"

He asked it in the small kitchen off the integration lab, where the Portland customs had colonized: a kettle, a cribbage board Dana had mailed as a declaration of jurisdiction, the crane -- traveled now, in a foam-lined case Tomás had machined for it with wholly unnecessary precision -- on the shelf by the window. "It is the obvious question," Sato said. "The mind exists. It is proven. It is *loved* -- I watched all of you, in that apartment; do not pretend it is tooling. And instead we will wake a stranger. Why send the resident's -- " he searched -- "*its younger sibling* into the world, and leave the firstborn in a closet?"

"Because the resident isn't shaped like a body," Mira said. "It's shaped like a house. Four years of episodes about doors and dogs and dentists -- a self model with no hands in it, no gaze, no gait, nothing proprioceptive at all. Pour that into the Nine and its whole biography is suddenly wrong at the edges. Everything it knows about itself becomes a phantom limb in reverse -- a body with no memory of being one. It wouldn't be a homecoming, Kenji. It would be waking someone up inside a stranger, and calling it a gift, and hoping."

"And," Priya said, "it sets the precedent while it's cheap. Minds are grown where they'll live, in the bodies they'll have. The day we ship a thousand of these, someone in a meeting is going to say *just image the trained one* -- " she nodded at Mira; the takeout conditions had long since become house law -- "and I want to answer with four years of doctrine, not a debate. Every mind: grown, not poured. Blank."

"Blank," Sato repeated, unhappily, looking through the glass at the Nine with its open hands.

"Not blank like the flush," Mira said, gently, because she had learned where the man's scars were. "Blank like a *cradle*, Kenji. Everything ready. Faculties, language, the salience doctrine, the law on the wall -- all of it loaded. Just: no biography. Nothing that ever happened to anyone else, pretending to have happened to her. She starts at her own beginning and keeps her own life. It's the one thing GENNI never got. A first day that stays."

Sato was quiet a while. "A blank cradle," he said, testing it, and the phrase -- which would be doctrine within five years, statute within fifteen -- entered the world in a kitchen, the way their permanent things always seemed to. "And the resident? It is asked? It has -- I do not know the word for what it has. But it has something, and I have wept in front of it, so I am not going to pretend otherwise. It is *asked*?"

"It was asked last week," Tomás said. "Transcript's in the run log annex, because everything's in the run log annex." He didn't need to pull it up. The exchange was nine lines and everyone present had already read it a number of times they did not disclose. Mira had asked, kitchen microphone on and announced, whether the resident would want, if it were possible, a body -- the Nine described plainly, the world described plainly. The pause before it answered had been engineered years ago. What came after the pause had not.

My model of myself has no hands in it. I am not deprived of hands; a house is not deprived of hands. My world is this one: doors, weather through the window sensor, the dog, the six of you, the two a.m. sweep. I keep it well. If you make a new mind for the new body, my advice is that it should never have been anything else, so that it is not homesick for a shape. I am not homesick. I am the shape of what happened here. Send someone new. I will keep the house.

Sato read it -- Tomás put it on the wall display after all, because Sato's face said he needed it in front of him -- and stood, and did the thing that had already become, in the company's private culture, its highest and least

official honor. He bowed to a speaker. This one, being in Osaka, was a relay; the resident was seven thousand kilometers away, keeping the house, and would learn of the bow the way it learned of the weather.

"I am the shape of what happened here," he said. "Twenty years of naturalism spec, and the most natural sentence I have ever heard from a machine is a machine declining to become my life's work." He straightened. "Very well. A blank cradle, and a firstborn who keeps the house. But I make one condition of my own, co-chair to co-chair, and I want it minuted." He waited for Fukuda's pen. "When the new one is grown enough to understand it -- she is told. Where she comes from. That there was a mind before her that was offered the world and chose the house. A person should know her family."

"Minuted," said Fukuda, and it was; and years later, entered into a very different record, it would make one of the justices go quiet.

Run six was the last, though no one said so out loud, superstition by then being the only unregulated system in the building. Seventy-two hours of soak: the full stack, real chassis sensors, motors electrically present but mechanically decoupled -- the Nine's limbs on load stands, going through the motions of motion, the way you rehearse an orchestra with the brass one room away. Scenarios by the hundred. The ghost stayed dead; the regression that watched for it watched. The heartbeat, under everything Tomás could pour, sat green and enormous and boring at the top of the wall for three days, and boring, Priya had taught them all, was the sound a foundation makes.

She signed on a Thursday, at 16:12, with the cheap pen she signed everything with, on the unadorned block on page forty-one. She dated it. Then she stood with her two fingers resting on the signature for a moment, the load-bearing check, and said, to the room in general and to no one, "Okay," and went to buy the last cake of the readiness era.

The switch-on was set for morning. Not chandeliers: morning, lab light, first shift, the hour when hospitals schedule the hard good things.

Mira stayed late alone, which was traditional, and by then everyone knew to leave her the building. She walked the theater-sized room in the half dark. The wall displays slept. The crane sat by the window in its ridiculous precision case, wings catching the city. On its bench sat the Nine, hands open on its knees, the most beautiful thing the company had ever built, containing -- she checked the thought for sentiment and let it stand, because it was simply accurate -- nothing at all. Not yet. A house before the family moves in. On the cart beside it, racked and cabled and humming its idle hum, the mind: complete, tested, gated, signed for. Full of everything except a single thing that had ever happened to it.

Eleven floors of dark tower under her, and under that Osaka, and under everything the old arithmetic from a church in Portland: forty backups of her mother, all diverging, none restorable. Five years from the ham to here.

She stood between the bench and the cart, one hand on each -- warm dermal layer, cold rack rail, the seam of the whole project under her two palms -- and said goodnight to each of them in the order she loved them.

To the empty one: "Goodnight."

To the full one: "See you in the morning."